Module Proforma

UNIVERSITY OF WESTMINSTER

Module full title	Software Engineering Final Project
SITS Module Code	6SENG010W
Credit level	Level 6
Length	Year Long
UK credit value	40
ECTS credit value	20
College	Design, Creative and Digital Industries
School	Computer Science and Engineering
Host programme of study	BECSS01F (BEng Software Engineering FT)
Progression and assessment board	COMENG - Computing UG PAB
Pre-requisites	
Co-requisites	
Study abroad	No
Special features	Franchising at IIT (Sri Lanka)
Access restrictions	
Are the module learning outcomes delivered, assessed or supported through an arrangement with organisation(s) other than the University of Westminster:	No
Summary of module content	This module requires students to integrate previously acquired knowledge and techniques with new insights gained through investigation and research, culminating in an extended piece of work. It involves the conceptualisation, design, implementation, and evaluation of a substantial solution, such as a software application, process, model, or experimental study. The module fosters independent working and encourages students to apply their learning to real-world challenges, supported by research, design thinking, documentation, and critical evaluation. Throughout the project lifecycle, students are expected to engage with ethical considerations and embed sustainability principles in a professional and autonomous manner.

Module aims

The module aims to develop your ability to work independently and consolidate knowledge gained throughout the course by undertaking a substantial project in the domain of software engineering. You will learn to transform a concept into a solution, either through rigorous research that compares approaches or by designing and implementing a solution to solve a problem. The module equips you with the skills, methodologies, and processes needed to address real-world challenges autonomously and professionally, while critically engaging with ethical considerations and integrating sustainability principles throughout the project lifecycle.

Assessment Methods

Rank	Assessment type	Assessment name	Weighting	Qualifying Mark	Qualifying set (where the minimum mark required applies across multiple assessments)
	Presentation	Project Proposal	15	30	
	Presentation	Interim Progress Demonstration	15	30	
	Project	Final Project and Report	50	30	
	Viva	Final Project Viva	20	30	

Synoptic assessment

n/a

Learning outcomes

By the end of the module, the successful student will be able to:

LO1 Analyse a significant software engineering problem, identifying key challenges, requirements, and constraints within its context.

LO2 Design and justify a comprehensive solution by selecting and applying appropriate techniques, methods, and technologies; evaluate its feasibility and potential sustainable impact.

LO3 Conduct independent research using a range of academic and professional sources, critically evaluate the literature and evidence, and synthesise findings to support the project rationale and solution development.

LO4 Plan, manage, and reflect on project activities effectively, demonstrating sound project management, autonomous working, and efficient use of time and resources.

LO5 Critically evaluate and address equality, diversity and inclusion (EDI), legal, social, ethical, and professional considerations relevant to the problem and its solution, including sustainability implications.

LO6 Develop a functioning technical or analytical solution that demonstrates the proposed approach, or provide a critical analysis of a defined problem and related research, employing an appropriate methodology and effectively using relevant tools, frameworks, and technologies while applying advanced technical knowledge and skills.

LO7 Critically evaluate and communicate the outcomes of the project by applying appropriate techniques, presenting findings clearly and professionally through written documentation, oral presentation, and viva voce examination, while defending design choices and reflecting on results and limitations.

Course outcomes the module contributes to

BEng Software Engineering L6.01, L6.04, L6.05, L6.06

Indicative syllabus content

- Developing a project idea.
- Background and literature review.
- Project planning and management.
- Project Lifecycle – choosing a suitable lifecycle or methodology for the project.
- Ethical, legal and sustainability considerations.
- Project requirements – capturing and modelling project requirements, qualities and constraints using appropriate research methods.
- Design - experimental design, application design, user interface, data visualisation, prototyping.
- Experimentation and implementation.
- Testing.
- Report writing, evidencing, documenting, critical evaluation, presentation and style.

Teaching and learning methods

The teaching and learning strategy for the Final Year Project module is designed to develop your independence, research capability, and practical skills, enabling you to apply the theoretical knowledge and techniques gained throughout your degree to deliver a substantial project. You will be paired with a supervisor who provides ongoing guidance and feedback. A list of available projects is published online, and you may select preferred options or propose your own in consultation with a staff member. Supervisor allocation is based on your preferences, the supervisor's expertise, and availability.
Once assigned a project, you are expected to meet regularly with your supervisor, maintaining a record of progress, meetings, tasks, and deadlines. These meetings allow you to discuss challenges, receive feedback, and plan next steps. A range of learning methods supports your progress, including independent study, workshops covering each stage of the project lifecycle, online materials, and supervisory meetings. These approaches help you refine your research, design, and technical development while ensuring steady progress.
Ethical, social, and sustainable considerations are embedded throughout the module. You will complete an ethics application reviewed institutionally to ensure your project aligns with the University's commitment to Equality, Diversity, and Inclusion (EDI) and complies with relevant standards.

Activity type	**Category**	**Student learning and teaching hours ***
Lecture	Scheduled	
Seminar	Scheduled	
Tutorial	Scheduled	
Project supervisor	Scheduled	12
Demonstration	Scheduled	
Practical classes and workshops	Scheduled	18
Supervised time in studio/workshop	Scheduled	
Fieldwork	Scheduled	
External visits	Scheduled	
Work based learning	Scheduled	
Scheduled online learning	Scheduled	
Other learning	Scheduled	
Total scheduled		30
Placement	Placement	
Independent study	Independent	370
Total student learning and teaching hours		400

* hours per activity type are indicative and subject to change

Assessment rationale: Why has this assessment been used for this module?

The assessment strategy for this module is designed to reflect the integrated nature of a substantial, independent project, enabling you to demonstrate a full range of academic, technical, and professional skills expected at Level 6. The four components - Project Proposal, Interim Progress Demonstration, Final Project Report, and Final Project Viva - are structured to assess the progressive development of the project from conception to completion.

These assessments are intentionally designed as authentic tasks, encouraging experiential learning and the application of critical thinking, creativity, and transferable skills to address real-world challenges. You are expected to engage in enquiry that extends beyond the classroom, developing capabilities relevant to your personal growth, employability, and civic responsibility.

From the initial stage of issue identification, you are required to consider the United Nations Sustainable Development Goals (SDGs) as a guiding framework, alongside ethical principles and Equality, Diversity and Inclusion (EDI) values. These considerations must be embedded throughout the project lifecycle, and you will be assessed on how effectively you integrate them into your research, design, implementation, and reflection.

This combination of written, practical, and oral assessments ensures that you are evaluated holistically, in ways that closely mirror professional and research practice in computing disciplines, while also fostering a strong sense of global

awareness and responsibility.

Project Proposal (PP) – Presentation (up to 20 minutes)

- Learning Outcomes: LO1, LO2, LO3, LO4, LO5
- The Project Proposal (PP) provides you with an early opportunity to demonstrate your ability to identify and articulate a viable and significant computing problem, and to undertake initial research and analysis to inform the direction of your final project. It allows you to define clear objectives, requirements, and a structured project plan, while considering sustainability and ethical implications. This assessment evaluates your ability to formulate and communicate a well-founded problem statement, justify the proposed solution through critical analysis, and effectively plan the project. The proposal serves as the foundation for the subsequent project work, ensuring feasibility, clarity of scope, and alignment with academic and professional standards.

Interim Progress Demonstration (IPD) – Presentation (up to 20 minutes)

- Learning Outcomes: LO2, LO4, LO5, LO6
- The Interim Progress Demonstration (IPD) provides a checkpoint to evaluate progress against the initial proposal. Through a presentation, you are required to demonstrate your evolving technical and analytical work, critically reflect on the development process, and provide evidence of engagement with your supervisor. This assessment is designed to test your ability to communicate technical progress, justify methodological choices, and demonstrate either the partial implementation of your proposed solution or the progression in the critical analysis of a defined problem and related research. It encourages reflective practice, adaptability, and the ability to manage a complex, evolving project, key professional competencies in the sector.

Final Project and Report (FPR) – Developed artefact and Written Report (10,000-12,000 words excluding code and diagrams)

- Learning Outcomes: LO2, LO3, LO5, LO6, LO7
- The Final Project and Report (FPR) is the main summative component, capturing the full project lifecycle. It assesses your ability to apply advanced technical and analytical skills to produce a coherent, rigorous, and professionally focused piece of work. It evaluates your developed solution by assessing how well it meets the project's requirements and addresses the defined problem. It provides evidence of independent investigation, critical analysis, integration of theory and practice, and consideration of ethical, legal, social, and sustainability factors, as well as reflective learning.

Final Project Viva (FPV) – Viva (up to 45 minutes)

- Learning Outcomes: LO2, LO7
- The Final Project Viva evaluates your ability to present and defend your work, demonstrating strong technical knowledge, a clear understanding of the project, justification of your decisions, and professional communication skills. It also assesses your critical thinking, academic integrity, and awareness of future development, while confirming the authenticity of the completed project.

Formative feedback is embedded throughout the project lifecycle to support your academic development and guide the refinement of your work. Each formative assessment is purposefully designed to prepare you for subsequent summative tasks, culminating in the successful delivery of your Final Project.

You are expected to demonstrate the progression of your project at regular intervals. Your assigned project supervisor will provide ongoing, constructive feedback during scheduled meetings, helping you to critically reflect on your work, address challenges, and enhance the quality of your output.

Assessment criteria: What criteria will be used to assess my work on this module?

University Grade Descriptors are a benchmark point of reference, they are contextualised using specific subject specialist criteria specific to a particular assessment.

University Grade Descriptors

Criteria for assessments are designed with reference to the University's generic criteria to measure students' ability to meet the learning outcomes of a module. In addition, specific grade criteria are provided below:

Project Proposal (PP):

- To what extent can the student demonstrate the ability to identify and articulate a viable and significant computing problem.
- To what extent can the student demonstrate the ability to conduct initial research and critical analysis to justify the problem and proposed solution.
- To what extent can the student demonstrate the ability to define clear objectives, requirements, and a structured project plan.
- To what extent can the student demonstrate the ability to integrate sustainability and ethical considerations from the outset.
- To what extent can the student demonstrate the ability to demonstrate presentation, planning, and analytical skills.

Interim Project Demonstration (IPD):

- To what extent can the student demonstrate the ability to communicate technical progress and justify methodological choices.
- To what extent can the student demonstrate the ability to reflect critically on development challenges and decisions.
- To what extent can the student demonstrate the ability to demonstrate partial implementation and technical competence.
- To what extent can the student demonstrate the ability to engage with feedback and supervision effectively.
- To what extent can the student demonstrate the ability to demonstrate professional presentation skills.

Final Project and Report (FPR):

- To what extent can the student demonstrate the ability to demonstrate independent investigation through a literature review and analysis of relevant tools, frameworks, and techniques.
- To what extent can the student demonstrate the ability to apply advanced technical and analytical skills to develop a clear methodology, design, and implementation.
- To what extent can the student demonstrate the ability to justify the selection of technologies and approaches used throughout the project.
- To what extent can the student demonstrate the ability to conduct appropriate testing and critically evaluate the effectiveness of the solution.
- To what extent can the student demonstrate the ability to reflect on personal learning, challenges faced, and project evolution.
- To what extent can the student demonstrate the ability to integrate EDI and sustainability considerations from problem definition to final evaluation, aligned with the SDGs.
- To what extent can the student demonstrate the ability to produce a coherent, well-structured, and professionally written report.
- To what extent can the student demonstrate the ability to implement a robust, professional, and efficient technical solution.
- To what extent can the student demonstrate the ability to apply appropriate technologies and methodologies to tackle the defined problem.
- To what extent can the student demonstrate the ability to demonstrate innovation, problem-solving, and technical proficiency.
- To what extent can the student demonstrate the ability to meet the requirements and critically evaluate the effectiveness of the solution.

Final Project Viva (FPV):

- To what extent can the student demonstrate the ability to present and defend the completed project with clarity and professionalism and demonstrate ownership of the work.
- To what extent can the student demonstrate the ability to justify design and implementation decisions under scrutiny.
- To what extent can the student demonstrate the ability to exhibit autonomy, confidence, and awareness of future directions.
- To what extent can the student demonstrate the ability to engage with feedback and supervision effectively.

The exact assessment criteria that will apply to each assessment will be stated in the assessment specification, which will

be available on Blackboard at the time of delivery.

How will this module enhance employability?

This module significantly enhances employability by enabling students to undertake a substantial, independent project that mirrors real-world professional practice. Through the development of a realistic product and accompanying documentation, students demonstrate their ability to manage and complete complex tasks, showcasing technical proficiency, problem-solving skills, analytical thinking, and project management abilities.

Students are encouraged to explore the job market and align their project with areas of personal or professional interest. The final project can be included in a professional portfolio and used to evidence domain knowledge and practical capabilities to potential employers. The module also supports the development of transferable skills, including communication, adaptability, and reflective practice.

By tackling a complex, authentic problem using a structured approach, students transition from academic study to professional readiness. They are assessed on their ability to identify and resolve challenges, justify decisions, and present their work formally: skills that are directly relevant to technical interviews and workplace scenarios.

Importantly, students are required to consider sustainability and the United Nations Sustainable Development Goals (SDGs) from the outset of their project. This ensures that their work reflects global awareness and responsibility, further strengthening their profile as future computing professionals.